MINOR PROJECT 1

LOAN APPROVAL PREDICTION SYSTEM

Artificial Intelligence and Machine Learning (AIML) Track

Submitted by	RAHUL KUMAR
Roll Number	301302224267
College	RUNGTA COLLEGE OF ENGINEERING AND TECHNOLOGY
Department / Branch	Computer Science and Engineering (CSE Core)
Academic Session	2024–2028
Faculty Guide	SHEKHAR KUMAR

GitHub: https://github.com/rahulkt2411/loan-approval-prediction-system

Live application: https://loan-approval-prediction-system-8dn6o5usdvqfd2y8tvmcjw.streamlit.app/

Declaration

I, RAHUL KUMAR, declare that this report describes my Minor Project 1 titled “Loan Approval Prediction System” for the AIML track at RUNGTA COLLEGE OF ENGINEERING AND TECHNOLOGY. The project links and screenshots are included for review. Dataset details and quantitative model-evaluation metrics are not asserted unless they are verified from the actual notebook output.

Student Signature: _____ Date: _____

Acknowledgement

I express my gratitude to RUNGTA COLLEGE OF ENGINEERING AND TECHNOLOGY for providing the opportunity to undertake this project. I acknowledge the open-source Python, scikit-learn, Streamlit, and joblib communities and the dataset provider used in the notebook. Faculty guide details may be added if applicable.

Abstract

This project presents LoanWise, a web-based loan approval prediction prototype built for academic demonstration. The application allows a user to enter applicant profile and loan details, request a prediction, and view a model-estimated approval probability. The supplied application screenshots show an applicant form, a prediction result of “Likely approved” with a displayed probability of 63.3% for the example inputs, and a separate monthly EMI calculator. The application interface identifies the probability as a model estimate and states that it is not a guarantee or actual bank decision. Logistic Regression is the stated model for this project. The displayed 63.3% is one example prediction, not a model accuracy score.

1. Introduction

Loan applications are assessed using applicant and loan-related attributes. Supervised machine learning can learn associations between historical labelled examples and outcomes, then estimate a class for a new record. This project packages a loan-status classifier into a Streamlit application to make the prediction workflow accessible through a browser.

The application is an educational prototype. Its outputs should not be used as the sole basis for real credit, financial, or lending decisions.

2. Problem Statement

To develop an interactive web application that accepts applicant and loan details, uses a Logistic Regression classification model to estimate loan approval status, and presents the prediction to the user.

3. Objectives

• Provide a simple browser-based interface for loan-related input fields.

• Use Logistic Regression as the stated classification approach.

• Load a persisted model artifact for prediction in the Streamlit application.

• Display the predicted outcome and, where available, a model-estimated probability.

• Provide an EMI calculator as a separate financial calculation utility.

• Document the project, verify application behavior, and report only measured evaluation results.

4. Scope and Technology Stack

The project scope covers model-based classification and an interactive demonstration interface. The supplied screenshots show navigation for Dashboard, Make a Prediction, EMI Calculator, Model Performance, Feature Importance, Prediction History, and About Project. Only behavior actually implemented in the current deployed version should be claimed in a final viva or report.

• Programming language: Python (version to confirm from project environment).

• Machine-learning model: Logistic Regression (as provided by the student).

• Web application: Streamlit.

• Model persistence: joblib model artifact (repository file: loan_approval_model.joblib).

• Development environment: Google Colab.

• Version control and source hosting: GitHub.

5. Dataset and Input Features

Dataset name, provider, licence, size, exact target column, feature list, missing-value treatment, categorical encoding, feature scaling, and train/test strategy must be copied from the actual Colab notebook. These details are intentionally not guessed in this report.

Item	Details to verify from notebook
Dataset title and source	[Enter exact title, URL and licence]
Records and columns	[Enter actual counts]
Target column / labels	[Enter exact target and class mapping]
Model input features	[Enter actual ordered feature names]
Preprocessing	[Describe only implemented steps]
Evaluation method	[Train/test split or cross-validation details]

6. Methodology

• Data inspection: review the dataset columns, target labels, missing values, and data quality.

• Data preparation: apply the preprocessing actually implemented in the notebook.

• Model training: train Logistic Regression on the prepared training data, if training occurs in the notebook.

• Evaluation: use the notebook's documented holdout or validation approach and record measured metrics.

• Model persistence: save and load the trained estimator or pipeline using joblib.

• Application inference: collect form values, construct input data in the expected feature schema, and request a prediction.

• Presentation: display the prediction and model-estimated probability when supported by the model.

• Additional utility: calculate a hypothetical loan EMI separately from the ML prediction.

7. System Architecture

Data source → preprocessing → Logistic Regression training → evaluation → saved model artifact → Streamlit form → prediction and probability display.

The EMI calculator is a separate component: loan principal, annual interest rate, and tenure are used to calculate an estimated monthly instalment, total payment, and total interest. This calculation is not a model prediction.

8. Application Screenshots

The following screenshots were supplied by the student from the deployed Streamlit application.

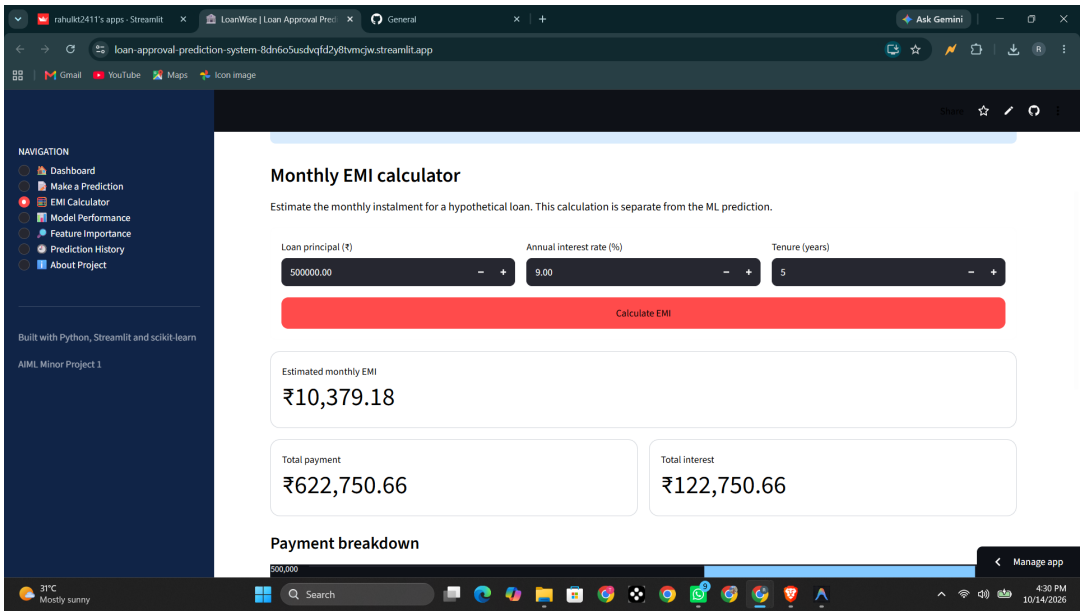


Figure 1. Applicant profile and loan details input form.

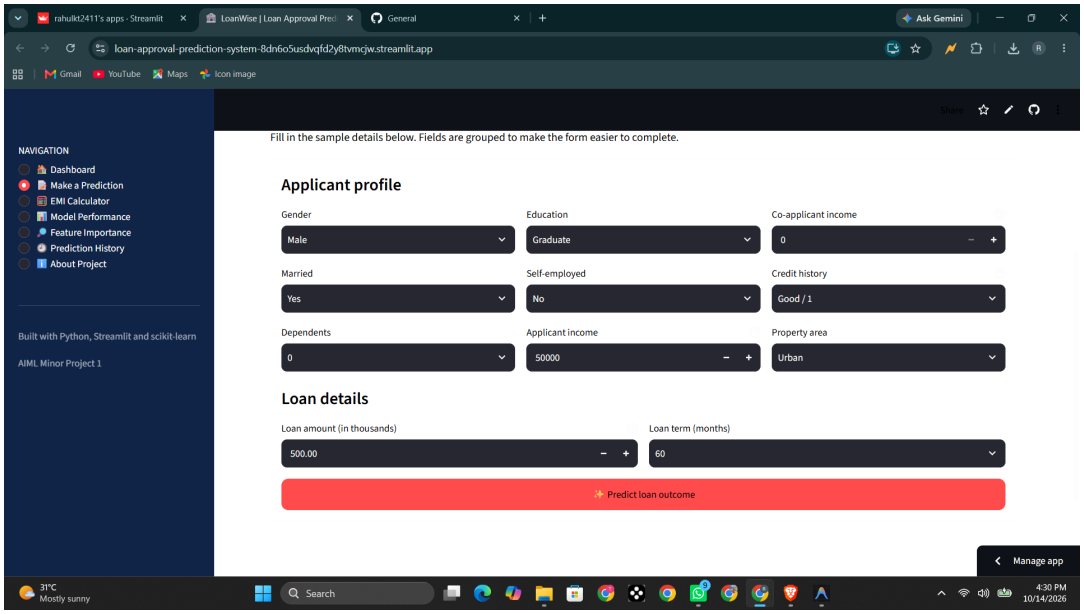


Figure 2. Example prediction result: “Likely approved” and 63.3% model-estimated approval probability.

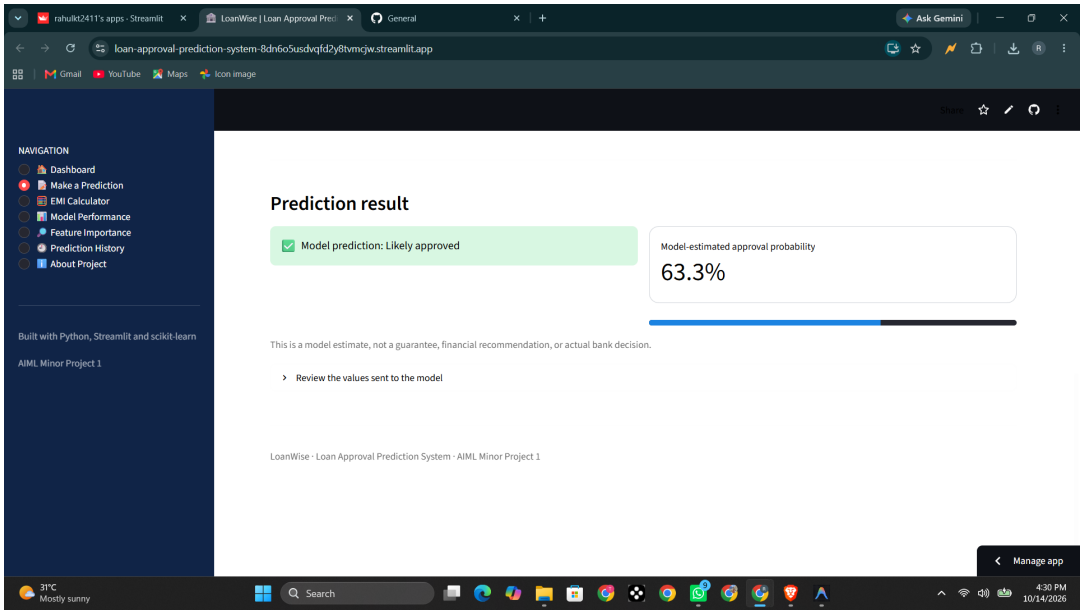


Figure 3. EMI calculator example showing principal ₹5,00,000, annual interest 9%, tenure 5 years, and estimated EMI ₹10,379.18.

9. Testing

Test case	Evidence / expected outcome	Status
Prediction form	Form fields visible in supplied screenshot	Observed in screenshot
Example prediction	Result and 63.3% probability displayed	Observed in screenshot
EMI calculator	EMI, total payment and interest displayed	Observed in screenshot
Model Performance / Feature Importance	Navigation entries visible; detailed behavior not verified	Verify live app
Model artifact loading	Repository includes a saved model artifact per prior repository review	Verify by running app
Invalid input handling	Not evidenced by supplied screenshots	Test before submission

10. Model Evaluation and Results

The supplied screenshot verifies one inference example, not overall model performance. A displayed approval probability of 63.3% must not be reported as 63.3% accuracy. The evaluation-results CSV was previously observed with metric headings but no populated result rows in its public preview. Therefore, this report does not state accuracy, precision, recall, F1-score, ROC-AUC, or a confusion matrix.

Before final submission, run the evaluation cells in the Colab notebook and record only metrics actually calculated on the stated test/validation data. Explain which label is considered the positive class and describe the split or validation method.

11. Limitations and Responsible Use

• Predictions depend on the training dataset, feature definitions, preprocessing, and representativeness of the examples.

• Historical lending data may contain bias; model performance should be assessed beyond overall accuracy.

• A predicted probability is a model output and may not be calibrated to real-world approval rates.

• Input preprocessing in the deployed app must match training-time preprocessing and feature ordering.

• The EMI calculator provides a mathematical estimate and may exclude fees, insurance, taxes, or lender-specific terms.

• This application is an academic prototype, not a real credit approval or financial advice system.

12. Conclusion and Future Scope

LoanWise demonstrates an end-to-end concept for presenting a loan approval classification model through a Streamlit interface. The supplied screenshots show the input form, an example prediction with a model-estimated probability, and a separate EMI calculator. Quantitative claims about predictive performance remain pending a verified evaluation run.

• Populate a reproducible evaluation report and confusion matrix.

• Validate inputs and add clear error messages for unsupported values.

• Review probability calibration and fairness limitations before any real-world interpretation.

• Add dataset attribution, versioned model training steps, and deployment instructions.

13. References

• Python documentation: https://docs.python.org/3/

• scikit-learn documentation: https://scikit-learn.org/stable/

• Streamlit documentation: https://docs.streamlit.io/

• GitHub repository: https://github.com/rahulkt2411/loan-approval-prediction-system

• Google Colab notebook: https://colab.research.google.com/drive/1HWvlstT-uvcoJwy5BrhV5cRekh0WQ8lC#scrollTo=P9UzhP-ZUfxz

• Dataset citation: [add the actual dataset provider, URL, licence, and access date].

Appendix A. Project Links

GitHub repository: https://github.com/rahulkt2411/loan-approval-prediction-system

Google Colab notebook:
https://colab.research.google.com/drive/1HWvlstT-uvcoJwy5BrhV5cRekh0WQ8lC#scrollTo=P9UzhP-ZUfxz

Live Streamlit application: https://loan-approval-prediction-system-8dn6o5usdvqfd2y8tvmcjw.streamlit.app/

Submission reminder: fill the dataset details and actual metrics, check the department/college's required format, then upload this report and PPT to the accessible repository or Drive folder.